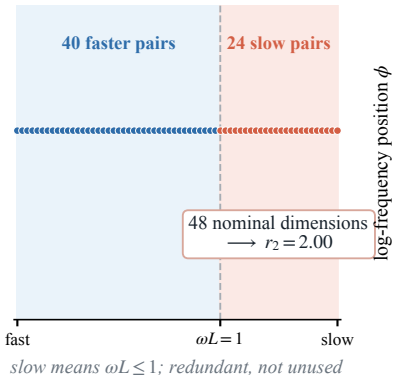
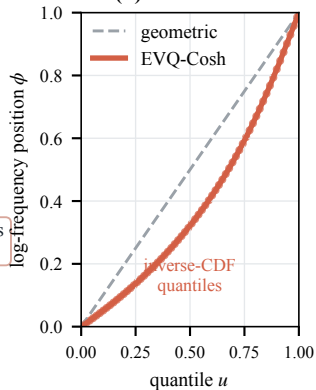


(a) Static low-frequency collapse



(b) Closed form



$$\phi_k = 1 - \tau^{-1} \operatorname{asinh}((1 - u_k) \sinh \tau)$$

(c) Identification and co-adaptation

FIX RANGE
same endpoints, span, init, data, budget
move only 30 interior frequencies
 $\Delta \text{OOD NLL} = -0.478 / -0.205 / -0.113$



CROSS WEIGHTS $\times$ RUNTIME TABLE
self-consistent PPL: 7.14 / 7.16
post-hoc swaps: 76.20 / 23.05

allocation is real; trained use is table-conditioned